

Naziru Halilu

Digital Agriculture, Computer Vision, Geospatial Machine Learning

Pamplona, Spain (originally Zaria, Nigeria)

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SUMMARY

Agricultural engineer with a long term research focus on the digitalization of agriculture, spanning computer vision, geospatial machine learning, agricultural control systems, and high throughput phenotyping, grounded in a strong foundation in agricultural machinery, mechanization, and farm power systems. First Class Honours B.Eng, CGPA 4.77 of 5.00, Top 1 percent. Currently an Erasmus Mundus Joint Master's scholar.

EDUCATION

Erasmus Mundus Joint Master's Degree (EMJMD), Sustainable Agriculture (SUSTAGRI)

2025 to 2027

Public University of Navarre, Spain; UTAD, Portugal; Agricultural University of Athens, Greece

B.Eng, Agricultural and Bioresources Engineering

2019 to 2024

Ahmadu Bello University, Zaria, Nigeria. First Class Honours. CGPA 4.77 of 5.00, Top 1 percent

National Diploma, Agricultural Engineering Technology

2016 to 2018

Samaru College of Agriculture, ABU. Distinction. CGPA 3.78 of 4.00, Top 1 percent

RESEARCH AND PROFESSIONAL EXPERIENCE

EMJMD Research Scholar, Agricultural University of Athens, Greece

Sep 2026 to Jan 2027

- Continued graduate level research training in smart design and management of irrigation systems.
- Built NaCROP, a modular Python pipeline with a Tkinter desktop GUI for evapotranspiration, soil water balance, irrigation scheduling, and a four step crop growth simulation for five smallholder crops around Zaria, Nigeria. [Code: GitHub](#)

Summer Intern, Computer Vision Group, Dept. of Computer Engineering, ABU, Zaria

Jun to Aug 2026

- Accomplished cross condition transfer of crop disease recognition models, as measured by 64.4 percent classification accuracy (macro AUC 0.882 against a 5.3 percent chance baseline) across 19 disease classes on 1,543 field images, by extending the Geometry Agnostic Contrastive Learning (GACL) framework with four new geometry invariant neural components. Formalized as a first author paper under review at Biosystems Engineering, Elsevier. [Code: GitHub](#)

EMJMD Research Scholar, UTAD, Portugal

Feb to Jun 2026

- Accomplished cross system precision agriculture prediction and input cost reduction, as measured by R squared between 0.978 and 0.995 for grape ripening and maize yield regressions and a 2.69 tonne cut in lime demand, by integrating GIS, remote sensing, and machine learning pipelines across more than 65 hectares of Portuguese trial sites spanning three contrasting cropping systems. [Code: GitHub](#)
- Accomplished more reliable soil nutrient and yield mapping, as measured by a roughly 20 percent reduction in residual kriging uncertainty across 17,864 samples on a 16.85 hectare maize field (Moran's I between 0.72 and 0.98, p less than 0.001), by developing Multitask Singularity Regression Kriging (MSRK). [Code: GitHub](#)
- Accomplished predictive harvest date forecasting for grape ripening, as measured by R squared of 0.9784 and a maturity forecast of 25.5 to 27 degrees Brix, by building an NDVI and logarithmic regression model from field must samples across five sampling dates at a six hectare UTAD vineyard, Quinta de Nossa Senhora de Lurdes, Vila Real. [Code: GitHub](#)
- Accomplished a costed, spatially targeted pest management programme, as measured by a sized pheromone diffuser plan of EUR 2,670.6 to 3,052.1 across 11 grape variety blocks, by running terrain characterisation and a 25 trap spatial monitoring network for grape moth (Lobesia botrana) across a 42.97 hectare vineyard estate in the Douro Demarcated Region. [Code: GitHub](#)

EMJMD Research Scholar, Public University of Navarre, Spain

Sep 2025 to Jan 2026

- Accomplished real time, field deployable crop health screening, as measured by 33.8 percent balanced accuracy against a 4.55 percent chance baseline across 22 disease classes on 1,543 validation images, by building a fourteen module open source pipeline for RGB and multispectral vegetation indexing, GLCM texture analysis, and CNN/Faster R-CNN inference. [Code: GitHub](#)
- Accomplished radiometrically consistent multi temporal vegetation monitoring, as measured by a 39 and 31 percent reduction in temporal NDVI variance for water and rooftop reference targets, by designing STADOS, a five stage adaptive dark object subtraction framework validated on real Sentinel 2 imagery over Portuguese corn parcels. [Code: GitHub](#)

- Accomplished accurate land suitability mapping for livestock slurry application, as measured by RMSE of 7.78 and R squared of 0.77 on a held out set of 127 sites, by building a two layer GIS and regression kriging pipeline across 550 spatially cross validated observations over 14.35 square kilometres near Tudela, Spain. [Code: GitHub](#)
- Accomplished a 47.8 percent reduction in annual soil loss and Nash Sutcliffe model efficiency between 0.88 and 0.94, by building a Bayesian grounded digital twin combining hydrological, hydraulic, and hybrid CNN-BiLSTM-attention sediment models, hold out R squared of 0.90. [Code: GitHub](#)

Undergraduate Project Assistantships, ABU, Nigeria

Mar 2024 to Mar 2025

- Contributed data collection, statistical analysis, and literature review across concurrent undergraduate research projects in water and irrigation systems, soil and erosion assessment, and postharvest and animal systems.

Project Assistant, Remote Data Lead, Smart Greenhouse Automation, IAR-ABU, Nigeria

2024 to 2025

- Engineered adaptive irrigation scheduling algorithms fusing real time soil moisture and microclimate telemetry for automated greenhouse control.

Student Research Assistant, Precision Crop Systems, NAERLS, Nigeria

2022 to 2024

- Trained and validated CNN crop stress detection models reaching 78 percent accuracy on multispectral UAV imagery, and automated UAV flight path planning.

Assistant Farm Manager, part time, Yankari Nasarawa Farms, Nigeria

2022 to 2024

- Managed automated climate control and water dosing systems for a commercial poultry facility.

Project Assistant, Smart Irrigation and Water Systems, BOMO Basin Project, Nigeria

2021 to 2022

- Implemented closed loop control logic for a one hectare smart drip irrigation array.

Research Intern, NCAM, Ilorin and IAR, Zaria, Nigeria

2017 to 2020

- Built Python data logging pipelines to benchmark mechanization efficiency, and supported baseline hydrological research.

PUBLICATIONS

- Halilu, N., and Sulaiman, J. (2026). Real Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. IJRSI, 13(7), 2951 to 2989.
- Halilu, N. (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. IJRSI, 13(7), 2920 to 2950.
- Halilu, N., and Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. IJRSI, 13(7), 2990 to 3020.
- Ismail, K., and Halilu, N. (2024). AI Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. Proc. AfCPA, 2(1), 100 to 120.
- Halilu, N., and Sulaiman, J. Geometry Agnostic Contrastive Learning (GACL). Biosystems Engineering, Elsevier. Under review.
- Halilu, N. (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (Ipomoea spp.). Environmental Monitoring and Assessment, Springer Nature. Under review.
- Halilu, N., and Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. European Journal of Agronomy, Elsevier. Under review.

TECHNICAL SKILLS

Machine Learning and AI	PyTorch, TensorFlow, Keras, scikit learn, CNNs
Geospatial AI and GIS	ENVI, QGIS, ArcGIS, Google Earth Engine, Sentinel imagery
Crop and Environmental Modelling	APSIM Next Generation, crop simulation
Programming and Data Science	Python, MATLAB, R, Pandas, NumPy, Git and GitHub, Zenodo

AWARDS AND HONORS

Erasmus Mundus Joint Master Scholarship, 2025. Best Graduating Student, B.Eng, 2024. Nigerian Federal Government Award for Academic Excellence, 2024. Best Undergraduate Research Paper Award, ABU, 2024. Best Overall Graduating Student, National Diploma, 2019. PTDF Undergraduate Fellow, 2023.

PROFESSIONAL MEMBERSHIPS

ASABE. Nigerian Institution of Agricultural Engineers (NIAE). Nigerian Society of Engineers (NSE). Pupil Engineer, COREN.